

Mohammad Sadegh Sirjani

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Education

Ph.D., Computer Science University of Texas at San Antonio, GPA: 3.93/4.0	Jan. 2025 - Present
B.Sc., Computer Engineering Ferdowsi University of Mashhad, Cumulative GPA: 3.01/4, Last 60 Credits: 3.89/4	Sep. 2018 - Feb. 2024

Research Interests

Internet of Things	Tiny AI	Edge AI
Embedded Systems	Energy Harvesting	Intermittent Computing

Research Experience

Research Assistant ASIC Laboratory, USA	Jan 2025 - Present
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- Developing energy-efficient neural network accelerators for edge AI applications on IoT devices.
- Investigating ultra-low-power ML inference using hardware-software co-design approaches.

Research Assistant Software Quality Laboratory, Iran	Aug. 2022 - Jan. 2024
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- Analyzed Java application execution traces using XRebel and JRebel for software enhancement.
- Developed code analysis techniques to identify class, interface, and enum connections.
- Built tool for recommending microservice migration strategies from monolithic codebases.

Research Assistant Web Technology Laboratory, Iran	Dec. 2023 – Aug. 2024
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- Designed database structure and web-based API for psychological survey data collection.
- Integrated LLMs to generate enhanced client assessment reports from survey data.
- Successfully deployed production application through iterative prompt engineering refinements.

Publications

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1. Nourbakhsh, E., **Sirjani, M. S.**, Mousavi, A., Nguyen, K., Quarles, J., Xie, M., & Slavin, R. (2026). “Are LLM Benchmarks Already Contaminated? A Systematic Review of Contamination Detection Methods”. *Preprint (accepted, GEM Workshop @ ACL 2026)*, <https://doi.org/10.13140/RG.2.2.12579.41764>.
 2. Mousavi, A., **Sirjani, M. S.**, Nourbakhsh, E., Xie, M., Slavin, R., Neely, L., Davis, J., & Quarles, J. (2026). “CogAdapt: Transferring Clinical ECG Foundation Models to Wearable Cognitive Load Assessment via Lead Adaptation”. *arXiv preprint arXiv:2605.22774*, <https://arxiv.org/abs/2605.22774>.
 3. Mousavi, A., **Sirjani, M. S.**, Nourbakhsh, E., Xie, M., Slavin, R., Neely, L., Davis, J., & Quarles, J. (2026). “MambaGaze: Bidirectional Mamba with Explicit Missing Data Modeling for Cognitive Load Assessment from Eye-Gaze Tracking Data”. *arXiv preprint arXiv:2605.22775*, <https://arxiv.org/abs/2605.22775>.

4. **Sirjani, M. S.**, Ahmad, M., Mousavi, A., Nourbakhsh, E. & Nguyen, K. (2026). "Optimizing Task Scheduling in Fog Computing with Deadline Awareness". In Proceedings of the *2026 IEEE 2nd International Conference on Secure IoT, Assured and Trusted Computing (SATC)*. IEEE. <https://doi.org/10.1109/SATC69565.2026.11542230>.
5. Ghaffari, A., Firuzi, V., Maleki, A., **Sirjani, M. S.**, & Abedini Bagha, M. (2025). "QTE-IoT: Q-Learning-Based Task Scheduling Scheme to Enhance Energy Consumption and QoS in IoT Environments". *Sustainable Computing: Informatics and Systems Journal*, <https://doi.org/10.1016/j.suscom.2025.101247>.
6. **Sirjani, M. S.**, Maleki, A., Pakmehr, A., Abedini Bagha, M., Ghaffari, A., & Pour Haji Kazem, A. A. (2025). "Controller placement in software-defined networks using reinforcement learning and metaheuristics", *Cluster Computing*, 28(10), 660, <https://doi.org/10.1007/s10586-025-05331-y>.
7. **Sirjani, M. S.**, Mousavi, S. A., & Sadeghi, M. (2024), "Data mining and cloud computing for customer pattern analysis and value maximization". In Proceedings of the *QICAR Conference* (pp. 339-344). IEEE. <https://doi.org/10.1109/QICAR61538.2024.10496623>
8. Mousavi, S. A., **Sirjani, M. S.**, Bozorg Zadeh Razavi, S. J., & Nikooghadam, M. (2023), "Sec-Vanet: Provably secure authentication protocol for sending emergency events in VANET". In Proceedings of the *IKT Conference* (pp. 86-91). IEEE. <https://doi.org/10.1109/IKT62039.2023.10433027>
9. Mousavi, S. A., Sadeghi, M., & **Sirjani, M. S.** (2023). "A comparative evaluation of machine learning algorithms for IDS in IoT networks". In Proceedings of the *IKT Conference* (pp. 168-174). IEEE. <https://doi.org/10.1109/IKT62039.2023.10433047>

Award & Honors

- Selected as a DAC Young Fellow – 63rd Design Automation Conference (DAC), 2026
- Fan Favorite Award – Draper Data Science Business Plan Competition, 2026
- Scholarship recipient – DAIR3 Data Science Summer School, 2026
- Winner of 2-Minute Video Presentation Contest at DAC 2025
- Selected as a DAC Young Fellow – 62nd Design Automation Conference (DAC), 2025

Professional Service

- Reviewer – Great Lakes Symposium on VLSI (GLSVLSI), 2026

Teaching Experience

- **Fundamentals of Operating Systems** (Fall 2025) – Dr. Sam Silvestro, UTSA
- **Data Science** (Summer 2025) – Dr. Amin Sahba, UTSA
- **Computer Organization** (Summer 2025) – Dr. Subhasish Das, UTSA
- **Fundamentals of Operating Systems** (Spring 2025) – Dr. Mimi Xie, UTSA
- **Fundamentals of Cloud Computing** (Spring 2024) – Dr. Somayeh Sobati-Moghadam, FUM
- **Principles of Compiler Design** (Fall 2023) – Dr. Haleh Amintoosi, FUM
- **Fundamentals of Data Mining** (Fall 2023) – Dr. Behshid Behkamal, FUM
- **Fundamentals of Cloud Computing** (Fall 2023) – Dr. Somayeh Sobati-Moghadam, FUM
- **Principles of Compiler Design** (Spring 2023) – Dr. Haleh Amintoosi, FUM
- **Principles of Web Development** (Spring 2023, Fall 2022) – Dr. Somayeh Sobati-Moghadam, FUM

Technical Skills

- **Programming Languages:** Python, C, C++, C#
- **ML/DL Frameworks:** TensorFlow, PyTorch, Keras, scikit-learn

- **IoT Protocols:** MQTT
- **Embedded Platforms:** NVIDIA Jetson (Orin Nano, Orin NX, AGX Orin, Nano, Xavier), Raspberry Pi 5, Arduino, STM32 ARM Cortex M4, MSP430FR5994
- **Tools & Technologies:** Docker, Kubernetes, Git, CUDA